HOSPITAL APPOINTMENT OPERATIONS

Real-World Excel Data Analyst Project — Project Brief

Role: Junior Data Analyst **Client:** MediCare Hospital Network **Tool:** Microsoft Excel only

Business Problem: Hospital management believes appointment capacity is not being used efficiently. There are concerns around no-shows, cancellations, long waiting times and uneven doctor/department workload. Management wants an Excel-based analysis that identifies the major operational bottlenecks and supports practical decisions.

What management wants to know

Area	Questions to answer in Excel
Appointments	How many appointments? How does demand change by month, weekday and time?
No-shows	Which departments, doctors and time slots have the highest no-show rate?
Cancellations	Where are cancellations concentrated?
Waiting time	Which departments/doctors have the longest waits?
Capacity	Which doctors appear underutilized or overloaded?
Revenue	What revenue is generated and what potential fee value is associated with missed appointments?

Excel project workflow

1. Raw Data → 2. Data Audit → 3. Cleaning → 4. Calculated Columns → 5. Pivot/Formula Analysis → 6. Charts → 7. KPI Cards → 8. Slicers → 9. Executive Dashboard → 10. Insights & Recommendations

Expected raw-data fields

Field	Use
Appointment ID	Unique record / duplicate check
Appointment Date & Time	Trend, weekday and time-slot analysis
Patient ID	Patient behavior
Doctor / Department	Doctor and department performance
City	Location analysis
Status	Completed / Cancelled / No Show
Wait Time	Patient experience
Fee	Revenue analysis
Patient Type	New vs Returning

Excel Analysis & Dashboard Plan

1. Data Cleaning

Check duplicates, blanks, inconsistent department/city names, date formats, invalid times, unreasonable waiting times and status-related errors. Keep the raw data unchanged and create a cleaned working table.

2. Calculated fields

Create useful fields such as Month, Weekday, Time Slot, Appointment Outcome, No-Show Flag, Cancellation Flag and Revenue-at-Risk Flag where appropriate. Use formulas that can be explained in an interview.

3. Analysis tools

Excel feature	Purpose
COUNTIFS / SUMIFS / AVERAGEIFS	Business metrics and filtered calculations
IF / IFS / AND / OR	Classification and flags
XLOOKUP	Lookup/reference information
Pivot Tables	Department, doctor, time and status analysis
Pivot Charts	Management-friendly visualization
Slicers	Interactive filtering

4. Executive Dashboard

KPI cards: Total Appointments • Completion Rate • No-Show Rate • Cancellation Rate • Average Wait Time • Total Revenue

Core visuals: Monthly Appointment Trend • Department Performance • Doctor Utilization • No-Show/Cancellation Analysis • Waiting-Time Analysis • Revenue Analysis

Filters: Date/Month • Department • Doctor • City • Patient Type • Status

5. Final business output

Problem found	Evidence from Excel	Recommended action
High no-show rate	Rate by department/doctor/time slot	Evaluate reminder and confirmation process
Long waiting time	Average/median wait by department/time	Review scheduling and peak-hour capacity
Underutilized capacity	Doctor utilization by period	Rebalance available appointment slots
Peak-hour congestion	Demand vs available capacity pattern	Shift capacity toward high-demand periods

Portfolio goal: Finish with one clean Excel workbook containing Raw Data, Clean Data, Analysis/Pivots and a professional Dashboard. The final story should clearly show **Problem** → **Analysis** → **Evidence** → **Recommendation**.

Important: Dashboard numbers and recommendations must be calculated from the actual dataset. The project brief intentionally does not provide fake results.